A 13-variable natural equation: source-only candidate

Nonnegative integer unknowns; 89 binary-operation abbreviations; exact degree 56 at the fixed ports.

The 13 unknowns below are distinct nonnegative integers:
`argJ, argFuel, argAlpha, argPellF, argIndex, argTime, argX`
`argScale, argOrdinate, argEta, argZeta, argAuxY, argGamma`
Each monospaced G-name below is an abbreviation, not an unknown. The fixed integers β, K, μ, ω are defined after the equation.

$\mathtt{G001} = \mathtt{argFuel} + 1$
$\mathtt{G002} = \mathtt{argAlpha} + 1$
$\mathtt{G003} = \mathtt{argIndex} + 1$
$\mathtt{G004} = \mathtt{argTime} + 1$
$\mathtt{G005} = \mathtt{argScale} + 1$
$\mathtt{G006} = \mathtt{argEta} + 1$
$\mathtt{G007} = \mathtt{argZeta} + 1$
$\mathtt{G008} = \mathtt{argJ} \cdot \beta$
$\mathtt{G009} = 1 + \mathtt{G008}$
$\mathtt{G010} = \mathtt{G009} \cdot \mathtt{G009}$
$\mathtt{G011} = \mathtt{G009} \cdot \mathtt{G010}$
$\mathtt{G012} = \mathtt{G011} \cdot \mathtt{G005}$
$\mathtt{G013} = \mathtt{G012} \cdot \mathtt{G010}$
$\mathtt{G014} = \mathtt{G009} - \mathtt{G001}$
$\mathtt{G015} = \mathtt{G014} - \mathtt{G002}$
$\mathtt{G016} = \mathtt{argX} \cdot \mathtt{G012}$
$\mathtt{G017} = \mathtt{G006} + \mathtt{G007}$
$\mathtt{G018} = \mathtt{G012} \cdot \mathtt{G017}$
$\mathtt{G019} = \mathtt{G006} + \mathtt{G018}$
$\mathtt{G020} = \mathtt{G019} \cdot \mathtt{G019}$
$\mathtt{G021} = \mathtt{G016} + \mathtt{G012}$
$\mathtt{G022} = 4 \cdot \mathtt{G021}$
$\mathtt{G023} = 3 + \mathtt{G022}$
$\mathtt{G024} = \mathtt{G021} \cdot \mathtt{G021}$
$\mathtt{G025} = \mathtt{G023} + \mathtt{G024}$
$\mathtt{G026} = \mathtt{G021} \cdot \mathtt{G019}$
$\mathtt{G027} = \mathtt{argX} + \mathtt{G026}$
$\mathtt{G028} = \mathtt{G016} \cdot \mathtt{G018}$
$\mathtt{G029} = \mathtt{argX} + K$
$\mathtt{G030} = \mathtt{G015} \cdot \mathtt{G029}$
$\mathtt{G031} = \mathtt{G030} - \mathtt{G001}$
$\mathtt{G032} = \mathtt{G028} + \mathtt{G017}$
$\mathtt{G033} = \mathtt{G032} + \mathtt{argOrdinate}$
$\mathtt{G034} = \mathtt{G017} \cdot \mathtt{G031}$
$\mathtt{G035} = \mathtt{G033} - \mathtt{G034}$
$\mathtt{G036} = \mathtt{G008} \cdot \mathtt{G035}$
$\mathtt{G037} = \mathtt{G036} - \mathtt{G031}$
$\mathtt{G038} = \mathtt{G037} \cdot \mathtt{G037}$
$\mathtt{G039} = \mathtt{G008} \cdot \mathtt{G008}$
$\mathtt{G040} = \mathtt{G028} \cdot \mathtt{G032}$
$\mathtt{G041} = 1 + \mathtt{G040}$
$\mathtt{G042} = \mathtt{G039} \cdot \mathtt{G041}$
$\mathtt{G043} = \mathtt{G038} - \mathtt{G042}$
$\mathtt{G044} = \mathtt{G014} \cdot \mathtt{G008}$
$\mathtt{G045} = \mathtt{G002} + \mathtt{G044}$
$\mathtt{G046} = \mathtt{G010} - 1$
$\mathtt{G047} = \mathtt{G045} \cdot \mathtt{G046}$
$\mathtt{G048} = \omega \cdot \mathtt{G009}$
$\mathtt{G049} = \mu + \mathtt{G048}$
$\mathtt{G050} = \mathtt{argJ} \cdot \mathtt{G049}$
$\mathtt{G051} = \mathtt{G047} + \mathtt{G050}$
$\mathtt{G052} = \mathtt{G017} - \mathtt{G051}$
$\mathtt{G053} = \mathtt{G052} - 1$
$\mathtt{G054} = \mathtt{G013} \cdot \mathtt{argGamma}$
$\mathtt{G055} = \mathtt{G016} \cdot \mathtt{G009}$
$\mathtt{G056} = \mathtt{G054} + \mathtt{G055}$
$\mathtt{G057} = \mathtt{G056} - \mathtt{G053}$
$\mathtt{G058} = \mathtt{G013} \cdot \mathtt{G027}$
$\mathtt{G059} = \mathtt{G023} \cdot \mathtt{G057}$
$\mathtt{G060} = \mathtt{G058} + \mathtt{G059}$
$\mathtt{G061} = \mathtt{G060} \cdot \mathtt{G060}$
$\mathtt{G062} = \mathtt{G013} \cdot \mathtt{G013}$
$\mathtt{G063} = \mathtt{G025} \cdot \mathtt{G020}$
$\mathtt{G064} = 1 + \mathtt{G063}$
$\mathtt{G065} = \mathtt{G062} \cdot \mathtt{G064}$
$\mathtt{G066} = \mathtt{G061} - \mathtt{G065}$
$\mathtt{G067} = \mathtt{argPellF} \cdot \mathtt{argPellF}$
$\mathtt{G068} = \mathtt{G067} - 1$
$\mathtt{G069} = \mathtt{G025} \cdot \mathtt{G068}$
$\mathtt{G070} = \mathtt{G020} \cdot \mathtt{G003}$
$\mathtt{G071} = \mathtt{G070} \cdot \mathtt{G070}$
$\mathtt{G072} = \mathtt{G071} - \mathtt{G069}$
$\mathtt{G073} = \mathtt{G004} \cdot \mathtt{G019}$
$\mathtt{G074} = \mathtt{G073} - \mathtt{G051}$
$\mathtt{G075} = \mathtt{G067} \cdot \mathtt{G074}$
$\mathtt{G076} = \mathtt{G075} - \mathtt{G019}$
$\mathtt{G077} = \mathtt{argAuxY} \cdot \mathtt{argAuxY}$
$\mathtt{G078} = \mathtt{G076} \cdot \mathtt{G076}$
$\mathtt{G079} = \mathtt{G077} - \mathtt{G078}$
$\mathtt{G080} = \mathtt{G069} \cdot \mathtt{G079}$
$\mathtt{G081} = \mathtt{G077} - \mathtt{G080}$
$\mathtt{G082} = \mathtt{G043} \cdot \mathtt{G043}$
$\mathtt{G083} = \mathtt{G066} \cdot \mathtt{G066}$
$\mathtt{G084} = \mathtt{G081} - 1$
$\mathtt{G085} = \mathtt{G084} \cdot \mathtt{G084}$
$\mathtt{G086} = \mathtt{G072} \cdot \mathtt{G072}$
$\mathtt{G087} = \mathtt{G082} + \mathtt{G083}$
$\mathtt{G088} = \mathtt{G085} + \mathtt{G086}$
$\mathtt{G089} = \mathtt{G087} + \mathtt{G088}$

The single equation is $\boxed{\mathtt{G089} = 0}$.
The zero-map to the positive parent is an external standard-integer argument; its nonstandard model transfer and iPRA endpoint remain open.

The four fixed integers. All numerals, including the complete seed below, use ordinary base-10 notation. Put $\rho = 2^{67}$, $\beta = \rho^{8807} - 1$, and let $d_j = \lfloor P/10^{16j} \rfloor \bmod 10^{16}$ for $0 \le j < 943$. Let the ordered list $(\lambda_0, \ldots, \lambda_{40})$ be

$$1023, 1024, 1025, 1026, 1027, 1028, 1029, 1030, 1013, 1014, 1015, 1016, 1017, 1018,$$
$$1019, 1020, 1021, 1003, 1004, 1005, 1006, 1007, 1008, 1009, 1010, 983, 984, 985,$$
$$986, 987, 988, 989, 990, 991, 1031, 1022, 1011, 992, 1095, 2195, 3296$$

Define

$$\begin{aligned} K &= \rho^{30} + \rho^{62} + \rho^{1100} + \rho^{2201} + \rho^{8849} \\ &\quad + \sum_{j=0}^{942} d_j (\rho^{4393-j} + \rho^{5495-j}) \\ &\quad + \sum_{j=0}^{40} 2^{10+j} (\rho^{4393-\lambda_j} + \rho^{5495-\lambda_j}). \end{aligned}$$

For integer intervals $[a, b] = \{a, a+1, \ldots, b\}$, let

$$\begin{aligned} S = (&[0, 942] \cup [983, 992] \cup [1003, 1011] \cup [1013, 1031] \\ &\cup \{1063, 1064, 1095, 2195, 3296\}) \setminus \{1\}. \end{aligned}$$

Put $\mu = \beta - \sum_{k\in S} \rho^k - 2\rho^{1063}$ and $\omega = \beta + H$, where

$$\begin{aligned} H = 4 + \sum_{k=1013}^{1022} \rho^k + \sum_{k=1065}^{1073} \rho^k + \rho^{2195} + \rho^{3296} \\ + (2^{51} - 2)(\rho^{4393} + \rho^{5495}) + (2^{65} - 1) \sum_{k=8792}^{8804} \rho^k + (2^{19} - 1)\rho^{8805}. \end{aligned}$$

Complete decimal integer P**.** Concatenate these lines in order, without spaces or line breaks:

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The compact fixed-port recipe is distinct from the current-2011 equation. The external parent review is for standard integers and does not establish an iPRA endpoint for this expression.